

BHANUKA JANAPPRIYA NAMBUWASAM

Computer Science Undergraduate | University of Peradeniya

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PROFESSIONAL SUMMARY

Motivated 4th-year Computer Science (Hons) undergraduate at the University of Peradeniya conducting active NLP research on low-resource language structural discovery. Experienced in building full-stack AI systems, RAG pipelines, computer vision applications, and embedded IoT solutions. Proficient in end-to-end product delivery from LLM integration and real-time WebSocket backends to hardware prototyping. Seeking a Software Engineering or QA Engineer internship to contribute to robust enterprise technology.

EDUCATION

BSc. (Hons) in Computer Science, University of Peradeniya *Expected Graduation: 2027*

Current GPA: 3.14 / 4.00

Major Subjects (Years 1 to 2): Chemistry • Computer Science • Statistics

G.C.E. Advanced Level Chemistry: B • Physics: B • Biology: C • Z-Score: 1.3742

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (Node.js), TypeScript, MySQL

AI, ML & NLP: LLMs, RAG Architecture, LangChain, ChromaDB, NLTK, TransferLearning, PyTorch, Scikit-learn

Web & Backend: MongoDB, Django, Django REST Framework, Django Channels, FastAPI, React.js, Tailwind CSS,

Computer Vision & Graphics: OpenGL, OpenCV, Lucas-Kanade Optical Flow, Chromaticity-Based Modeling

Embedded Systems & Hardware: Arduino, Embedded C++, Onshape, IoT, Sensor Integration, 3D Printing

Tools & Frameworks: Git, Docker, Electron, Ollama, Google Colab, Matplotlib, Linux/Terminal

Design & Marketing: Adobe Illustrator, Adobe Photoshop, Canva, Social Media Marketing

KEY TECHNICAL PROJECTS

PROSE: Probabilistic Framework for Structural Discovery **[Ongoing Research]**

Python • NLP • mBERT • ByT5 • LLMs • Sinhala NLP • MRR Evaluation

- Conducting active research on structural discovery in low-resource and ancient languages, utilizing Sinhala as validation proxy for the probabilistic framework.
- Evaluating model performance using MRR and Top-K metrics against mBERT and ByT5 baselines to benchmark low-resource language understanding.

SoulCare: Online Stress & Depression Detection Platform **[Ongoing]**

Django • Django Channels • React.js • TypeScript • Tailwind CSS • Llama 3.2 • Ollama • NLTK • WebSockets • Redis

- Built a full-stack mental health platform with role-based access (patient/doctor/counselor/admin), real-time WebSocket chat, stress detection games, digital assessments, and a JWT-secured community forum.
- Integrated a private local Llama 3.2 AI companion via Ollama with NLTK crisis-keyword detection, Text-to-Speech, and breathing exercises; all inference runs fully on-device for privacy.

RAG-Based Customer Support Chatbot

FastAPI • Streamlit • LangChain • ChromaDB • HuggingFace MiniLM-L6 • Llama 3.2 • Ollama • SSE Streaming • Docker

- Built a fully local RAG pipeline: PDF/Markdown ingestion, MiniLM-L6 embeddings, ChromaDB vector store with similarity-threshold filtering, and cited Llama 3.2 responses with 10-turn session memory.
- Exposed a token-streaming FastAPI SSE endpoint with source citations and Streamlit UI; wrote pytest suite mocking all external services; containerised with Docker Compose.

Financial AI Agent: Warren Buffett Style Analysis

[Python](#) · [Generative AI](#) · [LLMs](#) · [Financial Modeling](#)

- Automated qualitative and quantitative analysis of public companies (ROE, intrinsic valuation) via a scalable GenAI pipeline bridging financial theory with practical AI product development.

Desert Scene 3D Procedural Engine

[C++](#) · [OpenGL](#) · [3D Graphics](#) · [Procedural Generation](#)

- Custom graphics engine with sine/cosine terrain heightmaps, fractal vegetation, particle dust storms, and a full Day/Night lighting cycle including directional sun/moon, campfire flickering, and time-of-day fog density.

ChromaSkin Pro

[C++](#) · [Python](#) · [OpenCV](#) · [Computer Vision](#) · [Optical Flow](#)

- Skin region detection using chromaticity-based modeling and Lucas-Kanade optical flow; deterministic, fully modular architecture prioritising explainability over black-box ML approaches.

NER Pipeline comparison

[Python](#) · [Elmo](#) · [FastText](#) · [BiLSTM](#) · [PyTorch](#) · [NLTK](#)

- A comprehensive study of tokenization (BPE, WordPiece, NLTK) and embedding methods (Word2Vec, GloVe, FastText, ELMo, BERT) for Named Entity Recognition using BiLSTM-CRF.

Smart Water Tank Level Monitor **[Ongoing]**

[Arduino](#) · [IoT](#) · [Embedded C++](#) · [Sensor Integration](#)

- An IoT system engineered for real-time water level tracking and automated leak detection in residential or industrial water tanks, utilizing flow rate and ultrasonic sensors connected to a central microcontroller.

Low-Cost Turbidity Meter

[C++](#) · [Arduino](#) · [SolidWorks](#) · [Circuit Design](#) · [3D Printing](#)

- Led hardware implementation: MD0591/ADS sensor integration, Arduino C++ firmware for signal processing, and independently 3D-printed SolidWorks enclosure.

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

Champion | IEEE INSL Central Provincial New Innovation Competition (Hope Project) *2023*

Champion | KickStart Weekend, SIRED Business Idea Competition *2023*

Junior Treasurer | Computer Society of University of Peradeniya (CSUP) *Jan 2026 – Present*

Oversees financial records, budget planning, and HR coordination for society initiatives.

Public Relations Officer | CSUP *Jan 2025 – Feb 2026*

Managed digital presence and PR campaigns; grew society online engagement.

Social Media Manager | CSUP *Jun 2025 – Jan 2026*

Organizing Committee Member | DataEx, Data Science Society, UoP *2026*

Chairperson, Product Designing Workshop | WeLead Career Skills Society, UoP *Oct 2024 – Jan 2025*

Web Master & PR Officer | SIRED Society *2024 – 2025*

Managed digital presence and public relations for the university entrepreneurship society.

Team Leader | Intrafaculty Chess Championship *2023 - 2025* Led team to 1st Runners-up placement.

REFERENCES

Dr. Erunika O. Dayaratna

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